



LINEAR ALGEBRA AND ITS APPLICATIONS

Renna Sultana

Department of Science and Humanities

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION

Renna Sultana

Department of Science and Humanities

Course Content: Row Exchanges and Permutation Matrices

❖ Consider the system of equations $Ax=b$. While solving the system if a zero appears in the pivot position, it calls for a row exchange.

This row exchange is taken care by **Permutation Matrices P**.

Here $A \neq LU$ then $PA=LU$ where **P is a Permutation Matrix** which is an Identity Matrix with rows in different order.

Ex: Consider the system $y=b_1; 2x-3y=b_2$

$Ax = b \Rightarrow \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_1 \\ b_2 \end{pmatrix}$ Here Gaussian elimination fails and so calls for a row exchange i.e., $\begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$

This is same as $PAX=Pb$

$$PA = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} 0 & 1 \\ 2 & 3 \end{pmatrix} = \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} = U \text{ and } Pb = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$$

$$\therefore PAX = PB \Rightarrow \begin{pmatrix} 2 & 3 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = \begin{pmatrix} b_2 \\ b_1 \end{pmatrix}$$

Permutation Matrices:

- ❖ **P is a Permutation Matrix** which is an Identity Matrix with rows in different order.
- **Product of two permutation matrices is also a Permutation Matrix.**
- **Inverse of a permutation matrices is also a Permutation Matrix.**
- **P^{-1} is always same as P^T .**
- **Permutation Matrices of order 2 are $2!=2$ in number.**

$$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad P_{21} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

❖ Permutation Matrices of order 3 are $3!=6$ in number.

$$I = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} ; \quad P_{21} = \begin{pmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{pmatrix} = P_{21}^{-1} ;$$

$$P_{31} = \begin{pmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ \textcolor{blue}{1} & 0 & 0 \end{pmatrix} = P_{31}^{-1} ; \quad P_{32} = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} = P_{32}^{-1} ;$$

$$P_{21}P_{31} = \begin{pmatrix} 0 & 1 & 0 \\ 0 & 0 & 1 \\ 1 & 0 & 0 \end{pmatrix} = (P_{21}P_{32})^{-1} ; \quad P_{21}P_{32} = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 0 \\ 0 & 1 & 0 \end{pmatrix} = (P_{21}P_{32})^{-1}$$

LINEAR ALGEBRA AND ITS APPLICATIONS

MATRICES AND GAUSSIAN ELIMINATION:

$$\text{Ex: } A = \begin{pmatrix} 2 & 3 & 3 \\ 6 & 9 & 8 \\ 0 & 5 & 7 \end{pmatrix} \xrightarrow{R_2 - 3R_1} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 0 & -1 \\ 0 & 5 & 7 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = U$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 3 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} \neq A$$

$$P_{23}A = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 6 & 9 & 8 \\ 0 & 5 & 7 \end{pmatrix} = \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 6 & 9 & 8 \end{pmatrix} \xrightarrow{R_3 - 3R_1} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = U$$

$$LU = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 3 & 0 & 1 \end{pmatrix} \begin{pmatrix} 2 & 3 & 3 \\ 0 & 5 & 7 \\ 0 & 0 & -1 \end{pmatrix} = P_{23}A$$

Problem: Explain why A is not factorizable into LU? How can A be modified so that the new matrix can be factored into LU? Also obtain the factors L,D,U for the new matrix. What is the relation between L and U thus obtained? Explain.

$$A = \begin{pmatrix} 1 & -2 & 2 \\ 2 & -4 & 5 \\ -2 & 5 & -4 \end{pmatrix} \xrightarrow[R_3+2R_1]{R_2-2R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{pmatrix} \xrightarrow{R_2 \leftrightarrow R_3} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix}$$

A is not factorizable into LU as Gaussian elimination fails and Row exchange is required.

So A should be multiplied with permutation matrix P_{23} so that $PA=LU$.

$$P_{23}A = \begin{pmatrix} 1 & -2 & 2 \\ -2 & 5 & -4 \\ 2 & -4 & 5 \end{pmatrix} \xrightarrow[R_3-2R_1]{R_2+2R_1} \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} = U \quad L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix}$$

$$L = \begin{pmatrix} 1 & 0 & 0 \\ -2 & 1 & 0 \\ 2 & 0 & 1 \end{pmatrix} \quad D = \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad U = \begin{pmatrix} 1 & -2 & 2 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \quad L=U^T$$



THANK YOU

Renna Sultana

Department of Science and Humanities

rennasultana@pes.edu